

How digital marketing can use a smart phone app to improve any business's bottom line

Journal of Information Technology Teaching Cases
2022, Vol. 12(2) 244–249
© Association for Information Technology Trust 2022
Article reuse guidelines:
sagepub.com/journals-permissions
DOI: 10.1177/20438869221075638

Barbara A Manko 

Abstract

The marketing strategy of offering customer loyalty rewards has evolved with the use of technology. Instead of printing old-fashioned paper coupons, punch cards, and flyers to advertise special offers, customers can access their rewards through a customized app. The app functions as an all-in-one tool to attract customers, send offers, communicate specials, take orders, offer new products and services to existing customers, and so much more. During the restrictions of the COVID-19 lockdown, businesses found online communication soaring, with the ability to offer no-contact delivery of goods and services, and many businesses are keeping the technology they implemented during lock down because it is just good business. The use of application creation for digital marketing has the ability to supercharge sales by more effectively reaching the target audience and providing business analytics on users.

Keywords

Mobile applications, website design, food ordering, digital coupons, smartphone apps, digital advertising, technology customer connection, social influence

Introduction

When science fiction movies picture the future, they often show scenes of targeted marketing. A customer walks into a store, his identity is scanned via his facial features, and immediately, he is greeted by name and offers are suggested for him based on his prior purchases. Describing this concept still sounds a little far out and futuristic, until one realizes that we are literally living in that future now. Instead of the offers being displayed on monitors or shouted out (which works well visually for a Hollywood blockbuster but is not very practical), they are literally at our fingertips through our phones. Marketing strategies have evolved since the days of Green Stamps and punch cards, but consumers are still motivated by discounts and rewards, and using the marketing strategy of offering perks for being a loyal member has continuously proven to positively impact a business's profits.

Any business has the ability to develop an app that will customize the user's shopping, dining, or interaction with that company, and create a personalized experience that boosts superior customer service, offers individualized rewards or incentives, and increases profit potential. Kaye Media Partners has a thought-provoking quote on their blog that says, "Generating engaging ideas is quite a challenge,

especially if you're playing in a fairly saturated market or have a product or service that doesn't lend itself to a multitude of topics. I mean how many ways can you talk about a light bulb?" But, think about this for a second: how many times have you gone to the store, holding the dead bulb in your hand, trying to figure out which weird size is going to match the style, wattage, and shape of the other three bulbs in the light above the dining room table or over the bathroom mirror. What if your neighborhood lightbulb store had an app that stored your past purchases, allowed you to scan the information on a light bulb, or even let you take a picture that would be automatically matched with possible options to help you reorder? What if it then led you to the correct aisle and shelf in the store with a handy map? And allowed you to pay through Apple pay and bypass even the self-checkout? Suddenly, the solution to a frustrating experience we have all shared becomes the thing

Gannon University, Erie, PA, USA

Corresponding author:

Barbara A Manko, Gannon University, 109 University Square, Office # 313, Erie, PA 16541, USA.
Email: manko001@gannon.edu

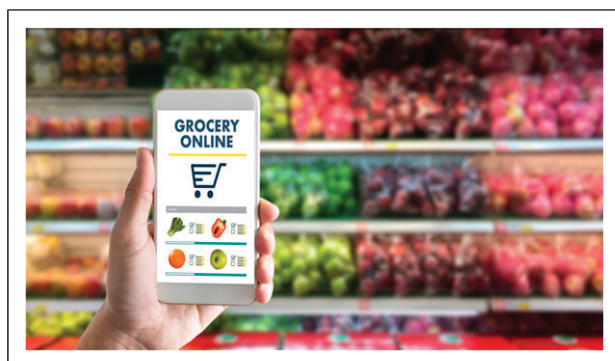


Figure 1. Image produced by ilanshatz on pixabay. Used by permission.

that sets this lightbulb store apart from the competition (Figures 1 and 2).

Apps for major chains have taken off in recent years. They exist for big box stores, grocery stores, hair salons, shopping boutiques, and restaurants. Rather than clipping coupons, like the old days, consumers can digitally add them and automatically apply them to their purchase at checkout. When thinking about the way people shop, “the use of tools, such as mobile coupons, has risen considerably compared to other communication and promotional tools. In addition, mobile coupons help retailers to send personalised promotional contents and marketing messages to consumers based on their location” (Carranza et al., 2020). Petit et al. (2019) analyze the rise in online purchases and what it could mean for the future. “Other sensory interfaces (e.g., including touch screens, together with a range of virtual, and augmented solutions) are increasingly being made available to people to interact online. Moreover, recent progress in the field of human–computer interaction means that online environments will likely engage more of the senses and become more connected with offline environments in the coming years. This expansion will likely coincide with an increasing engagement with the consumer’s more emotional senses, namely touch/haptics, and possibly even olfaction.” The possibilities are endless.

Consumers surveyed mentioned that “intangible attributes” guide their decision making when it comes to their business and extras such as the availability of Wi-Fi and other users’ reviews and ratings may become the deciding factor (Petit et al., 2019). For the sake of this article, the focus will mainly be on restaurant-related apps. With COVID-19 restrictions in place, many restaurants developed QR codes for customers to scan to access the menu rather than having paper menus. Online ordering ensures faster service or bypasses human interaction completely. A restaurant app allows access to the menu, specials, deals, and rewards. Using the app can lead to customer loyalty as diners earn points toward free food—appetizers, desserts,

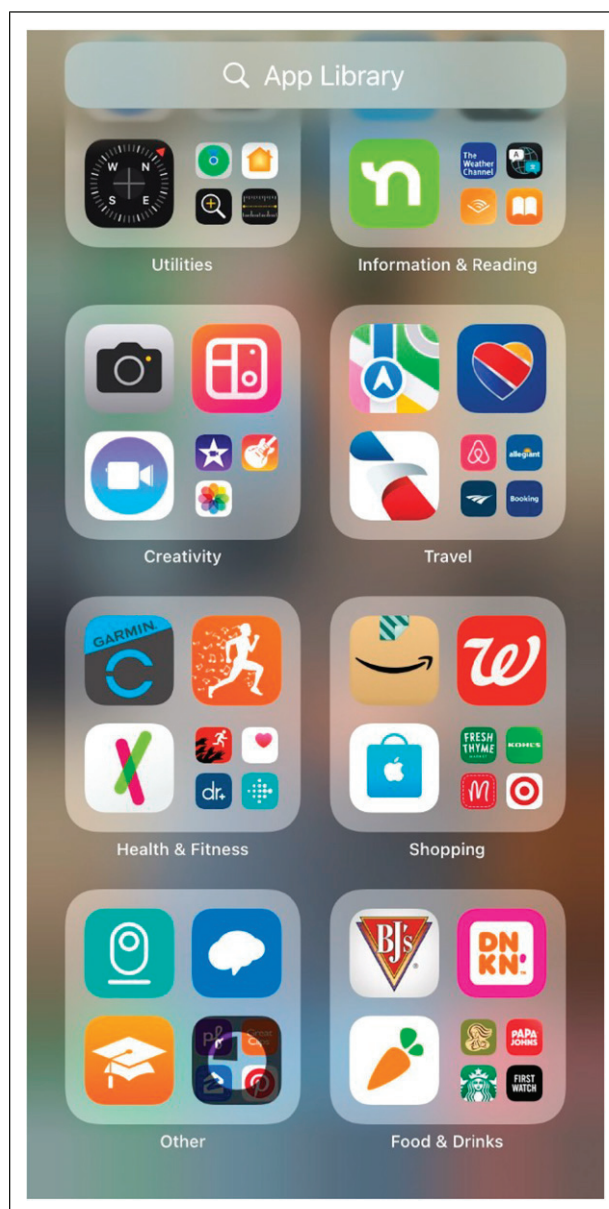


Figure 2. Phone interface. Image by the author.

and entrees. Aaron Allen, founder of restaurant consulting firm Aaron Allen & Associates, comments, “We’ve come a long way from the Subway punch card.”

Case study: Ordering from your favorite restaurant during a lockdown

The effects of COVID-19 will prove to be more far-reaching than anyone could have anticipated. Some are big and obvious, but some have fundamentally changed the way that businesses operate in small ways, such as the aforementioned discontinuance of a paper menu. Elle Kross, director

of client strategy for travel, hospitality and food services at digital marketing firm Movable Ink, told CNBC that there has been “a giant uptick” in the number of companies that “are now launching programs to keep consumers returning and keep that loyalty.” Chipotle CEO Brian Niccol shared that expansion to their loyalty program has increased members from 8.5 million members to 21 million members. They have emphasized the importance of the program by offering early access to new menu items and keeping some in-demand selections, such as the quesadilla, only available by digital order (Lucas 2021). As a result, digital orders made up the majority of the company’s sales for the first quarter of 2021 (ibid.). Loyalty programs are the perfect place to gather consumer data for analysis, allowing the company to better address its customers’ demands, and they directly lead to “more frequent visits and higher average checks,” says Allen (ibid.).

What many restaurants and their customers have found during 2020 is that diners were eager for access to their favorite meals, even if they could not eat inside the restaurant. Moreover, people enjoy the convenience of ordering online and being able to pick up dinner from the parking lot. The ease of reordering with an app that stores not only users’ credit card information but also their past and favorite orders, and automatically applies discounts and suggests add-ons like drinks and desserts, is not something that many people are eager to give up once things “return to normal.” Having indoor and outdoor seating, as well as curbside pickup, increases the number of customers that can be served in a shift, and as capacity limits return to normal and people revert to their normal busy schedules, taking advantage of the convenience of an app to order dinner ahead of time for quick pick-up between school and soccer is sure to be irresistible. The question is how businesses can optimize their profit margins while incorporating the new ways of doing things with the old.

Review of literature

Kukanov and Andrianova (2018) call the growth of the mobile app market “unprecedented” and explain that users are often choosing smartphones over computers or laptops. This has shifted the way that people think in many ways, including how they spend money. Consumers expect to have information readily available at their fingertips. Kukanov and Andrianova point out that “taking this into consideration [when] making the consumer applications navigator is needed by ordinary people.” An app is more user friendly for the sizing of the phone, and typically easier to navigate than the full website. “As mobile devices become more prevalent, mobile applications (or *apps*) become increasingly important to our daily lives, providing access to information, communication, and entertainment. Rather than using Web browsers as the main application to access

content, which applies to most desktop users, mobile apps help optimize access to specific content through customized UI designs. This is a necessity due to smaller screen sizes and limited resources on mobile platforms. Users would like apps to respond quickly to their requests, consume minimal mobile data to minimize their monthly bill, and consume less energy to ensure sufficient battery life” (Mao 2016).

Certain restaurant chains have introduced a tablet at the table where customers can order drinks or dessert, request the bill, and even play games (both free and paid versions). The replacement of the old triangle peg game has gone high tech. But the hardware investment does not have to be as high if you assume that every customer carries their own interface with them through their smartphone. One app can replace multiple marketing platforms: a rewards program, targeted advertising, information blasts to announce seasonal specials, menu changes, deals, etc., and a one-stop shop for reservations, orders, communication with a server, paying, and feedback.

Okumus et al. (2018) point out that many diners are already consulting their smart phone in a restaurant to figure out nutritional information and enter calories and other diet information into their fitness apps. Alon et al. (2020) estimate there are approximately 325,000 mobile health apps available for download. This can help diners with allergies or special requirements. “The restaurant industry can no longer ignore such demands for healthy eating environments. Many customers perceive restaurants to be socially responsible when they are provided with nutrition information and food information and they have favorable attitude toward healthy food options and they have high willingness to visit restaurants that provide healthy food options and nutrition information. These businesses can use smartphone apps to promote their business, menu items, specials, and ingredients. Smart apps can also inform their patrons about composition of ingredients, such as organic, non-GMO, vegan, vegetarian, pork free, gluten free, or other special diets.” Having nutritional information clearly displayed and easily accessed can be the deciding factor in which restaurant to frequent, whether the diner is trying to lose weight, does not eat red meat, or cannot be exposed to shellfish because of an allergy. Health tracking apps fit into a category called “digital therapeutic products” but they tend to “do little more than track activities such as sleep or exercise.” However, Loughheed explains that the possibilities extend much farther than simple spreadsheet tracking, and in reality, “software... can complement or even replace prescription drugs for managing certain health conditions” (2019). Having ingredient and nutritional information on the website, and in a form that can be easily translated to other apps, is a perk that many diners appreciate.

An important concept when it comes to the apps that people choose is *personalization*. “Personalization might be much more effective on social media compared to other

channels and suggests that this result could be due to the interactive feature of social media and its intrinsic personal nature. Similarly, it is proposed that through the use of conversational agents such as chatbots, firms will be able to better engage consumers through the use of personalized information” (Nobile and Kalbaska 2020). The five key factors of personalization are “offerings, knowledge, channels, purpose and contextual factors” (ibid.). Having the ability to customize the app can give the user a feeling of “perceived personalization” even if their actual experience is very similar to other users.

The blog from MJINFO has an article entitled “Digital Marketing Strategies that Would Boost Your Business Profit” that analyzes how a website drives profitability. They highlight the essentials for successful digital marketing strategies a good interface with speedy website loading, employing website analytics and SEO optimization, and offering unique content and optimizing social media to build brand awareness. By focusing on these categories when developing an app, a company may avoid some bugs the first time around.

Challenges to app development

There are two major challenges: for the business itself to have the knowledge of technology to create an effective interface and for customers to have the ability to access it. There remain some Luddites who simply do not own a smart phone, or even a computer. Going completely digital may not be feasible, and attempting to do so means having a backup plan for old-fashioned diners who cannot use a smartphone to see the menu or for instances when, God forbid, the Wi-Fi goes down. This should not be a major stumbling block, but it is something to consider when trying to make the transition.

“An enormous market for mobile app quality and user experience measurement tools has grown around the mobile app store ecosystems,” explain Braun et al. (2017). With the fast-paced development of the market and the needs of users, any company might struggle to keep up with the newest offering. “Currently there are alone more than 28 tools and frameworks for functional test automation and more than 16 different device clouds available. For most of the software developing companies, it is impossible to evaluate and test all of them” (ibid.). Forward-thinking marketers and researchers will therefore need to find a way to appropriate the latest tools/technologies in order to deliver richer online experiences for tomorrow’s consumers (Petit et al., 2019). Having a great app idea is only as effective as the development of that application for users across various platforms. But there is help for businesses out there!

Karen McFarlane of Kaye Media Partners specializes in working with clients to create marketing campaigns that include mobile marketing and social media. She says, “My practice specializes in targeting marketers and IT professionals which opens the doors to numerous types of clients

including cloud applications, SaaS/PaaS firms, mobile apps and other innovators in the space. My mantra is ‘Great Marketing Rocks’ and whether we work with the C-Suite, marketing teams, sales divisions or if we’re lucky – a combination of them all, our goal is to make our clients achieve rock star status making my job not only ripe with opportunity, but loads of fun.” Having the right resources in one’s pocket is key, and if they have passion for the project and the company, it can lead to success for all parties.

As with many aspects of business, there are platforms and third-party developers available to create the content needed for a successful app, which allows businesses of all sizes the ability to be competitive. “SMBs, particularly start-ups, need automation tools to grow their business at a more sustainable and rapid pace in order to compete in the marketplace” according to Kaye Media Partners’ blog article “3 Marketing Automation Platforms for SMBs.” They point out that focusing time and resources in the right places allows businesses to provide better customer experiences and expand. HubSpot offers website development templates that promise customization and individualization of carefully curated bundles designed to help a business with its social media needs. They promise “a website that’s attractive, user-friendly, and chock full of persuasive content.”

Beyond the interaction with current customers, the development of the right kind of website promises lead information and growth of the customer base. Kaye Media Partners sum up their strategy as “It’s all about staying in touch in a meaningful way and providing killer content to boot. The trickiest part is finding out what customers want to know and trying not to get on their nerves with endless sales pitches.” It is a delicate line to walk, and one that many businesses fail at, but when successful, and with the ability to automate the process, the far-reaching effects of this technique can translate into dollars for a business. Another option is Marketing 36[®], which promises “everything you need to build an amazing online storefront that can help grow your business” and offers tools to “create a sleek, modern online store, stay organized while managing customers and payments, sell your products everywhere online.” For businesses that are having difficulty transitioning to the use of technology, there are resources to guide them into the 21st century and beyond.

Conclusion

Developing an app offers as much to the business as it does to the customer. With the right third-party developer, the process should not be intimidating, and any business can benefit in a tangible way from implementing a user-friendly interface. It is just one prong on the digital marketing wheel that leads to a holistic marketing experience, but one that has seen impressive growth recently and promises the possibility of positive corresponding growth in financials.

Questions

1. App development and design are important in the creation of any mobile app, but the marketing aspect must also be considered, as students identify how marketing concepts can be utilized over a digital platform. In the steady stream of information available through a user's phone, how do developers ensure their app is being chosen, interacted with, and used?
2. The class may brainstorm what kind of rewards and perks motivate them to frequent one establishment over another, and how these may be incorporated into a digital format. Is there a place in the future for rewards programs and discounts? Should they be one-size-fits-all or can they be customized to the individual user?
3. Students may consider what features should be highlighted for quick access and how to organize the menu to make the experience intuitive and useful. Using their own experiences, have them plot a user-friendly app and identify features that make one clunky or difficult to navigate.
4. This article has focused mainly on the restaurant industry, but the possibilities are endless when it comes to apps. Have students scroll through their own phones and make a list of the relevant apps they have. Discuss which they actually use and if there are any they wish existed but do not at this time. Think about which apps allow direct payment and what they could mean in making a transition to a cashless society.

Acknowledgements

The author would like to acknowledge the chief editors from the Journal of Information Technology Teaching Cases for giving the author the opportunity to share knowledge and work experience with people from around the world. The author would like to also thank the anonymous reviewers for their insightful suggestions regarding the manuscript. Although the work on this paper was self-funded, the author wishes to thank the institutions that have employed her and the colleagues she has met through them. Dr George G. Couch, Assistant Professor, Healthcare Management and Assistant Professor, Business Administration; Dr Bruce A. Kibler, Assistant Professor, Management and Assistant Professor, Graduate Business Studies; Dr Michael J. Messina, Professor, Marketing Program and Professor, Graduate Business Studies; Dr Philip I. Szmedra, Assistant Professor, Business Administration and Graduate Business Studies; Dr Jinhee Yoo, Associate Professor, Management; Dr Terry S. Holmes, Assistant Professor, Accounting Program; and Dr M. G. Roth, Assistant Professor, Finance and Economics.

Declaration of conflicting interests

The author(s) declared no potential conflicts of interest with respect to the research, authorship, and/or publication of this article.

Funding

The author(s) received no financial support for the research, authorship, and/or publication of this article.

ORCID iD

Barbara A Manko  <https://orcid.org/0000-0001-6916-5717>

References

- Alon N, Stern AD and Torous J (2020). Assessing the food and drug administration's risk-based framework for software precertification with top health apps in the united states: quality improvement study. *JMIR mHealth and uHealth* 8(10): e20482. DOI: [10.2196/20482](https://doi.org/10.2196/20482).
- Braun S, Elberzhager F and Holl K (2017). Automation support for mobile app quality assurance - a tool landscape. *Procedia Computer Science* 110: 117–124. DOI: [10.1016/j.procs.2017.06.129](https://doi.org/10.1016/j.procs.2017.06.129).
- Carranza R, Díaz E, Martín-Consuegra D, et al. (2020). PLS-SEM in business promotion strategies. a multigroup analysis of mobile coupon users using MICOM. *Industrial Management & Data Systems* 120(12): 2349–2374. DOI: [10.1108/IMDS-12-2019-0726](https://doi.org/10.1108/IMDS-12-2019-0726).
- Kukanov A and Andrianova E (2018). *Creating the Mobile Apps Independence Rating*. Les Ulis, France: EDP Sciences. <http://dx.doi.org.ezaccess.libraries.psu.edu/10.1051/itmconf/20181803002>.
- Lougheed T (2019). How “digital therapeutics” differ from traditional health and wellness apps. *Canadian Medical Association Journal*, 191(43), E1200–E1201. DOI: [10.1503/cmaj.1095801](https://doi.org/10.1503/cmaj.1095801).
- Lucas A (2021). *Beyond the Punch Card: Restaurants Lean into Loyalty Program to Hold onto Pandemic Digital Gains*. “Tech to Table.” CNBC. 11 June 2021. <https://www.cnbc.com/2021/06/11/restaurants-lean-into-loyalty-programs-to-hold-onto-digital-gains.html> (accessed on 27 June 2021)
- Mao ZM (2016). Diagnosing mobile apps' quality of experience: challenges and promising directions. *IEEE Internet Computing*, 20(1), 66–69. DOI: [10.1109/MIC.2016.1](https://doi.org/10.1109/MIC.2016.1).
- Nobile TH and Kalbaska N (2020). *An Exploration of Personalization in Digital Communication. Insights in Fashion*. New York, NY: Springer International Publishing. 456–473. DOI: [10.1007/978-3-030-50341-3_35](https://doi.org/10.1007/978-3-030-50341-3_35).
- Okumus B, Ali F, Bilgihan A, et al. (2018). Psychological factors influencing customers' acceptance of smartphone diet apps when ordering food at restaurants. *International Journal of Hospitality Management* 72: 67–77. DOI: [10.1016/j.ijhm.2018.01.001](https://doi.org/10.1016/j.ijhm.2018.01.001).
- Petit O, Velasco C and Spence C (2019). Digital sensory marketing: Integrating new technologies into multisensory online experience. *Journal of Interactive Marketing*, 45, 42–61. DOI: [10.1016/j.intmar.2018.07.004](https://doi.org/10.1016/j.intmar.2018.07.004).

Author Biography

Barbara A. Manko's interest in the fields of Business, Marketing, and Information Systems and Technology led

her to pursue a Doctorate of Science in Information Systems and Commutation after completing her MBA. She has taught university courses in digital marketing, human resources, business analytics, strategic planning, and computer science for both undergraduate and MBA programs. Her recent research, published in 2020 and 2021, focuses on

digital marketing, technology management, analytics, online learning for students, and how online university programs prepare students to work remotely. Her upcoming research teaching case study will be published soon in the scientific journal "Management." The title is "The Pros and Cons of Telemarketing and its place in the business world."